

Precision-AMR: Spatially Adaptive Floating-Point Precision for PDE-Based Reverse Time Migration

A new axis of numerical adaptivity for adjoint wave-equation solvers, developed on dedicated NVIDIA hardware and validated across heterogeneous NVIDIA and AMD GPU architectures on the Santos Dumont supercomputer (LNCC)

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Abstract

Scientific simulations such as Reverse Time Migration (RTM) solve partial differential equations (PDEs) demanding FP64/FP32 accuracy, yet modern GPU architectures increasingly optimize for low-precision AI workloads (FP8/BF16) and expose new memory hierarchies, tensor cores, and mixed-precision math libraries. Prior work on RTM acceleration has reduced floating-point precision uniformly across the whole computational domain; separately, adaptive mesh refinement (AMR) has been explored for seismic wave modeling to vary grid spacing according to local wavelength. This project proposes a third, largely unexplored axis of adaptivity: Precision-AMR (P-AMR), a spatially block-structured scheme that varies floating-point precision — not mesh resolution — according to physically motivated, per-block error-tolerance criteria (proximity to absorbing boundaries, wavefield energy density, adjoint-sensitivity magnitude, and a time-dependent band trailing the propagating wavefront).

Unlike mesh-based AMR, where numerical artifacts are localized at coarse-fine grid interfaces, precision-induced round-off noise propagates with the wave itself, so P-AMR's tagging criteria must target where injected error will not reach the imaging condition or the FWI gradient, and must control round-off accumulation over the thousands of time steps typical of RTM/FWI. We build P-AMR on our prior GPUZIP framework, which accelerated RTM on V100 systems via GPU-side lossy compression (nvCOMP/Bitcomp) and checkpoint prefetching, cutting host-GPU transfer volume by up to $80\times$ and delivering up to $5.1\times$ end-to-end speedup. Differential checkpointing, cuBLAS/cuSOLVER mixed-precision linear algebra, Ozaki-scheme FP64 emulation, and cuFFT-based spectral operators are retained as supporting techniques, now applied selectively, per tile, according to the criteria that P-AMR establishes, rather than uniformly across the domain.

The 12-month programme combines two computational resources. An RTX PRO 6000 Blackwell GPU, requested as dedicated local hardware, supports day-to-day development and the selective-emulation study central to Phase 3, since its negligible native FP64 throughput makes the precision-zoning question unavoidable rather than optional. The Santos Dumont supercomputer (LNCC/SINAPAD) — whose partitions span NVIDIA Volta V100, Hopper H100,

and Grace-Hopper GH200 accelerators alongside AMD Instinct MI300A (CDNA3) APUs — supports the large-scale validation sweep across datasets, checkpointing algorithms, and P-AMR tagging criteria, and lets us test directly whether the tagging methodology generalizes beyond a single GPU vendor. Outcomes include an open-source GPUZIP v3.0 release built around a Precision-AMR module, and a peer-reviewed publication.

Keywords: reverse time migration; adaptive precision; mixed-precision computing; floating-point arithmetic; adjoint PDE solvers; block-structured adaptivity; cross-architecture portability; GPU computing; lossy compression; checkpointing.

1 Computational Platforms

HPC SDK (nvc++/nvfortran, CUDA-X math libraries, Nsight Systems/Compute). Experience: team uses CUDA and Nsight Systems routinely. Adoption: migrate the Awave-3D build from clang to nvc++ and validate numerical parity against existing baselines. Nsight Compute provides kernel-level roofline analysis, now applied per spatial tile, to identify which tiles are memory-bandwidth-bound versus compute-bound and to objectively scope where a lower-precision tile actually pays off.

RTX PRO 6000 Blackwell (dedicated hardware, requested). This GPU’s 5th-generation tensor cores with FP8/FP4 support and negligible native FP64 throughput make it the central platform for P-AMR: because full-domain FP64 emulation would be expensive everywhere, the scientific question this project answers is precisely which tiles need it and which do not. Its 96 GB GDDR7 ECC memory also supports local development; as dedicated, queue-free hardware it enables fast iterative debugging that a shared batch system is not well suited for.

Santos Dumont (LNCC/SINAPAD, Tier-0 national supercomputer, Petrópolis-RJ). Provides the large-scale validation environment. Its BullSequana X1000 partition includes 94 nodes with $4 \times$ NVIDIA Volta V100 GPUs each — matching the hardware used for the original GPUZIP baselines — plus a dedicated AI node with $8 \times$ V100 16GB over NVLink. Its newer BullSequana XH3000 partition adds 62 nodes with $4 \times$ NVIDIA Hopper H100 SXM 80GB over NVLink, 36 NVIDIA Grace-Hopper GH200 superchip nodes, and 18 nodes with $2 \times$ AMD Instinct MI300A APUs (CDNA3 GPU cores, 128 GB HBM3 each). This heterogeneity lets us generate FP64 ground truth and run the full P-AMR criteria sweep at scale on NVIDIA hardware, and separately test whether the tagging methodology and block-tiling approach generalize to a non-NVIDIA architecture.

nvCOMP / Bitcomp (GPU-side lossy compressor). Experience: integrated and validated in GPUZIP v2.0, where Bitcomp delivered the best accuracy-compression balance on V100. Adoption here: the storage mechanism for differential checkpointing, now indexed by the same tile identifiers used for precision tagging, so each tile carries a single, jointly-tuned precision-and-compression error budget.

cuBLAS (mixed-precision extensions) and cuSOLVER (iterative refinement). Both ship inside the HPC SDK. Experience: team has used BLAS/LAPACK routines in prior HPC work. Adoption: applied only within tiles that P-AMR flags as containing dense sub-kernels needing accuracy beyond native low precision (FP32 compute with FP64 accumulation, and cuSOLVER iterative refinement).

Ozaki-scheme FP64 emulation via low-precision tensor cores. Adoption: reserved for the subset of tiles that P-AMR’s sensitivity criteria mark as accuracy-critical (near the wavefront, high adjoint-sensitivity magnitude), rather than applied uniformly — directly testing whether selective emulation is cheaper than emulating everywhere while preserving image fidelity. Recent work has demonstrated up to $2.3\times$ speedup over native FP64 DGEMM on the RTX PRO 6000 Blackwell [5].

cuFFT. Experience: used in prior spectral operator work. Adoption: benchmarked as an alternative to FDM stencils for wavefield updates within selected, precision-tagged tiles, characterising the memory-bandwidth versus compute trade-off relative to the standard finite-difference path.

AMD Instinct MI300A (CDNA3, Santos Dumont, exploratory). As an additional validation activity, we assess how much of the P-AMR framework — the block/tile decomposition, the four tagging criteria, and the joint precision-compression error budget — transfers to AMD’s ROCm/HIP ecosystem (rocBLAS, hipBLAS, and AMD’s own reduced-precision matrix paths), without assuming feature parity with the CUDA-specific Ozaki-scheme emulation or nvCOMP/Bitcomp compression path. The goal is to determine whether Precision-AMR is a portable methodology or an NVIDIA-specific optimization.

2 Dataset and Model

- Size: $\approx X$ TB total (to be confirmed from REDU). Checkpoint pools reached ≈ 330 GB host RAM on V100; a single Marmousi3D checkpoint is ≈ 500 MB.
- Data type: seismic velocity models and traces (numerical grids).
- Source: Salt (SEG/EAGE) and Marmousi3D are open datasets; the Large dataset is in-house.
- Confidentiality: non-confidential, no personal data, no NDA.
- AI model parameters: not applicable (numerical PDE solver).
- Storage location: Universidade Estadual de Campinas (UNICAMP), Campinas, Brazil; staged for upload to Santos Dumont’s parallel file systems for large-scale runs.

3 Introduction

Hardware is trending toward low precision for AI, but PDE-based scientific simulation requires high precision to remain trustworthy. RTM, a seismic imaging method that solves the wave equation by finite differences over thousands of time steps, is a canonical example: geophysicists use the migrated image to guide expensive drilling decisions, so numerical fidelity is non-negotiable. RTM is also memory-intensive, requiring hundreds of gigabytes of wavefield state and relying on checkpointing to trade memory against recomputation in its forward/backward adjoint structure.

Our prior work addressed the dominant bottleneck — host-GPU checkpoint transfer over PCIe — with GPUZIP [3, 4], which compresses checkpoints on the GPU and prefetches them proactively, reaching up to $5.1\times$ speedup on V100 while preserving image quality (SSIM ≈ 1 , near-zero relative error). Separately, reduced-precision arithmetic has already been shown to be viable for seismic modeling, RTM, and FWI: Fabien-Ouellet [1] found that a stable half-precision (FP16) implementation of the elastic wave equation is achievable through appropriate scaling, that the resulting seismogram error is a negligible fraction of total seismic energy and largely incoherent with seismic phases, and that this noise does not adversely affect FWI or

RTM image quality. However, that reduction is applied uniformly across the entire domain. Gao and Harms [2] identified a specific hazard of that approach: naive half-precision time stepping accumulates round-off error through a disguised recursive summation in the update rule, degrading solution quality, with compensated (Kahan) summation as an effective remedy.

At the same time, adaptive mesh refinement (AMR) has been explored for seismic wave simulation and, more recently, explicitly for FWI and RTM workflows, to avoid the waste inherent in a uniform grid: dispersion and dissipation errors require grid spacing proportional to the shortest wavelength, so a uniform mesh is forced to use the finest spacing everywhere, even where the local velocity field would tolerate a much coarser one. AMR addresses this by refining Δx locally. No comparable, physically-motivated scheme exists for floating-point precision: existing mixed-precision work in seismic imaging is either global and uniform (as above), or adaptive only at the level of an individual linear-algebra operation (mixed-precision GEMM, iterative refinement, Ozaki-scheme emulation), not tied to a spatial decomposition of the wavefield itself.

This project proposes to close that gap with Precision-AMR (P-AMR): an adaptive-precision analogue of AMR that keeps mesh resolution fixed and instead varies the floating-point format assigned to each block (tile) of the domain, based on criteria that predict whether the round-off error introduced in that tile can reach the imaging condition or the FWI gradient. This is a structurally different problem from mesh-based AMR. In AMR, the characteristic artifact is a localized reflection at the coarse-fine grid interface, because refinement changes the discretization itself. In P-AMR, no discretization boundary is introduced; instead, a spatially varying level of round-off noise is injected into a hyperbolic PDE, and that noise propagates together with the wave. Tagging criteria therefore cannot rely solely on local smoothness (as AMR refinement indicators do); they must instead approximate where an error, once injected, will decay, be absorbed, or otherwise fail to reach the receivers or the sensitivity kernel before the simulation ends.

We propose four candidate criteria, detailed in Methods: proximity to the absorbing boundary (PML), where energy is already being discarded on purpose; local wavefield energy density, which governs how much a given tile’s error can contribute to the final correlation; adjoint-sensitivity magnitude, reusing the gradient computation that FWI already performs; and a time-dependent band trailing the propagating wavefront, analogous to a moving shock detector in astrophysical or combustion AMR codes. Because RTM/FWI already require a block/tile decomposition of the domain for checkpointing, P-AMR reuses that same decomposition as its tagging granularity, unifying the precision budget and the lossy-compression budget of differential checkpointing into a single, jointly tuned per-tile accuracy target. Mixed-precision GEMM, cuSOLVER iterative refinement, and Ozaki-scheme emulation remain essential, but are now applied selectively, only where P-AMR’s criteria say they are needed, which is the natural way to exploit GPUs such as the RTX PRO 6000 Blackwell that offer negligible native FP64 throughput.

Because the tagging criteria are motivated by the physics of wave propagation and adjoint sensitivity rather than by any particular vendor’s arithmetic units, we treat their portability as an explicit question rather than an assumption. Access to the Santos Dumont supercomputer — whose partitions include AMD Instinct MI300A accelerators alongside several NVIDIA generations — lets us test that question directly, rather than merely claim it.

4 Methods

4.1 Phase 1 — Infrastructure and Baseline (M1–M2)

Port Awave-3D and GPUZIP to the HPC SDK (nvc++), re-integrate nvCOMP/Bitcomp, and reproduce V100 baselines on Santos Dumont’s V100 partition, matching the original GPUZIP hardware.

- Establish a block/tile decomposition of the domain, extending the existing checkpoint-block structure so that it can also carry a per-tile precision tag.
- Profile every kernel with Nsight Systems and Nsight Compute, on both the RTX PRO 6000 and Santos Dumont, confirming that FDM stencils are memory-bandwidth-bound at the tile granularity before any precision changes are attempted.

4.2 Phase 2 — Precision-AMR: Criteria, Tagging, and Validation (M2–M6)

This is the core scientific phase, run at scale on Santos Dumont’s H100 partition. We design, implement, and validate four candidate tagging criteria for spatial precision zoning:

- **Boundary criterion:** tiles inside or adjacent to the PML/absorbing layer are tagged for reduced precision by default, since their contribution is already being damped.
- **Energy criterion:** tiles are tagged by a threshold on local wavefield energy density, so that regions of low amplitude (before wavefront arrival, or after geometrical spreading has attenuated the signal) tolerate lower precision.
- **Sensitivity criterion (FWI):** tiles are tagged using the magnitude of the local adjoint-sensitivity kernel, reusing the gradient computation itself as the tagging signal, in direct analogy to a Berger–Oliger truncation-error estimator in classical AMR.
- **Wavefront-trailing criterion:** a time-dependent band immediately behind the propagating wavefront is tagged for reduced precision, while the leading edge — where phase accuracy governs the imaging condition — retains full precision.

For each criterion, we implement compensated (Kahan) summation in the low-precision tiles’ time-stepping update to control round-off accumulation over the full simulation, following the mechanism identified by Gao and Harms [2]. We explicitly test for a “precision-interface” artifact — analogous to coarse-fine grid-interface reflections in mesh-based AMR — via residual analysis at tile boundaries between differing precisions. Finally, each tile is assigned jointly a floating-point format and a checkpoint compression policy (full snapshot versus compressed delta), with combined accuracy measured by PSNR, SSIM, and mean relative error against the FP64 baseline, on both the raw wavefield and the final migrated image. This phase subsumes differential checkpointing, which becomes the storage mechanism for P-AMR-tagged tiles rather than a separate deliverable.

4.3 Phase 3 — Precision-AMR without Native FP64 (M7–M10)

The tagging criteria validated in Phase 2 are applied on the RTX PRO 6000, hardware with negligible native FP64 throughput. Per tile, the pipeline selects between native low-precision tensor-core execution, cuBLAS mixed-precision GEMM (FP32 compute with FP64 accumulation) or cuSOLVER iterative refinement for dense sub-kernels, and Ozaki-scheme FP64 emulation for tiles flagged as accuracy-critical.

- This directly operationalises the motivation for exploiting AI-first GPUs that lack native FP64: expensive emulation is concentrated only where P-AMR’s criteria say it is needed.
- We quantify the speedup/accuracy trade-off relative to a uniform-Ozaki baseline (emulating FP64 everywhere) to determine how much selective tiling actually saves.
- All errors are bounded jointly (compression $\times$ precision interaction) against the FP64 baseline using PSNR, SSIM, and mean relative error, across the same three datasets and three checkpointing algorithms (Revolve, zCut, Uniform) used throughout the project.

- Exploratory cross-architecture activity: the same tagging and tiling logic is ported to Santos Dumont’s AMD Instinct MI300A nodes via ROCm/HIP, to test whether the criteria and error-budget framework hold on a CDNA3 architecture without native access to Ozaki-scheme emulation or nvCOMP/Bitcomp.

4.4 Phase 4 — Spectral Operators, Integration, and Release (M10–M12)

- Benchmark cuFFT-based pseudo-spectral derivative operators as an alternative update rule for selected, precision-tagged tiles, as a secondary technique complementing the core P-AMR contribution.
- Integrate the Precision-AMR module, differential checkpointing, and mixed-precision paths into a unified GPUZIP v3.0 release.
- Multi-node scaling evaluation on Santos Dumont (≤ 8 GPUs).
- Final analysis, write-up, and submission, with Precision-AMR as the central scientific claim and differential checkpointing, mixed-precision linear algebra, and spectral operators presented as supporting techniques.

5 Evaluation

All experiments span three datasets (Large, Marmousi3D, Salt), three checkpointing algorithms (Revolve, zCut, Uniform), and the four P-AMR tagging criteria (individually and in combination). Kernel-level profiling uses Nsight Compute (roofline, memory-traffic breakdowns); timeline profiling uses Nsight Systems. Implementation is in C/C++ with CUDA under the HPC SDK, with OpenMP Cluster for multi-node scaling. Accuracy is evaluated on both the wavefield and the final RTM image, since a tiling that preserves wavefield PSNR/SSIM may still, in principle, shift the migrated image if error is spatially correlated with reflectors. For the cross-architecture activity, the same accuracy metrics are computed on Santos Dumont’s AMD MI300A partition and compared against the NVIDIA baselines, to characterise how much of the achieved speedup and accuracy trade-off is architecture-independent.

6 Expected Results

Timeline (12 months):

Month	Activity	Hardware	Phase
M1–M2	Port Awake-3D + GPUZIP to HPC SDK (nvc++); reproduce V100 baselines; establish block/tile decomposition; Nsight Compute roofline profiling per tile.	RTX PRO 6000; Santos Dumont (V100)	1
M2–M5	Design and implement Precision-AMR tagging criteria (PML proximity, energy density, adjoint sensitivity, wavefront-trailing band); compensated summation; precision-interface artifact testing.	Santos Dumont (H100)	2
M5–M6	Unify precision and compression error budget via differential checkpointing; multi-node scaling; open-source GPUZIP v3.0-alpha release; reproducibility package (REDU).	Santos Dumont (H100, multi-node)	2

Month	Activity	Hardware	Phase
M7–M10	Apply Precision-AMR tiling without native FP64: selective cuBLAS mixed-precision GEMM, cuSOLVER iterative refinement, and Ozaki-scheme FP64 emulation only in accuracy-critical tiles. Exploratory port of the tagging framework to AMD Instinct MI300A via ROCm/HIP.	RTX PRO 6000; Santos Dumont (AMD MI300A, exploratory)	3
M10–M12	cuFFT spectral-operator benchmark on selected tiles; unified GPUZIP v3.0 release; analysis, write-up, and paper submission.	RTX PRO 6000; Santos Dumont	4

Projected outcomes: an open-source GPUZIP v3.0 release (MIT license, GitHub) built around a Precision-AMR module, with differential checkpointing, mixed-precision GEMM/Ozaki-scheme paths, and cuFFT spectral operators integrated as supporting components; a peer-reviewed publication with Precision-AMR as the central contribution (target: SBAC-PAD / Euro-Par / IJHPCA); and a public reproducibility package (datasets, Nsight profiles, quality metrics) via UNICAMP REDU.

Dependencies: this hardware request is not conditional on being combined with another grant. Large-scale validation runs on the Santos Dumont supercomputer (LNCC/SINAPAD) through LNCC’s own computational allocation program (SINAPAD), independent of this request. Existing institutional support funds personnel; none of these sources duplicates the RTX PRO 6000 request.

7 Project Support Details

We request one RTX PRO 6000 Blackwell GPU as dedicated local development hardware. Its abundant FP8/FP4 tensor-core throughput and negligible native FP64 make it the scientifically relevant target for the selective-emulation study central to Phase 3: it forces the question of whether Precision-AMR’s selective use of Ozaki-scheme emulation can match full-domain emulation at a fraction of the cost. As dedicated, queue-free hardware, it also supports the iterative day-to-day development that a shared batch system is not well suited for. Large-scale validation — the P-AMR criteria sweep across datasets and checkpointing algorithms, FP64 baseline generation, multi-GPU scaling, and the AMD MI300A portability test — runs separately on the Santos Dumont supercomputer (LNCC/SINAPAD), accessed through LNCC’s own allocation program and outside the scope of this request.

Resource Request

Resource	Request
Physical hardware	1 × RTX PRO 6000 Blackwell — dedicated local development hardware, used throughout the project

Compute time on Santos Dumont is requested separately through LNCC’s own project allocation program (SINAPAD) and is not part of this hardware request.

References

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